ReviewLens: Sentiment Classification of E-Commerce Product Reviews

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Purpose

Online sellers receive thousands of product reviews, and finding unhappy customers by hand is slow. We will build a system that reads a review and classifies its sentiment as **negative, neutral, or positive**, so sellers can quickly find negative reviews and respond to problems. We will solve this one problem with three different methods (a fine-tuned pretrained transformer, a prompted large language model, and a large language model fine-tuned with QLoRA) and compare their performance on the same data.

Dataset

Amazon Reviews 2023 (McAuley Lab, UC San Diego): https://amazon-reviews-2023.github.io/main.html. We will use only the *Cell Phones & Accessories* category. We will first keep reviews containing 100-300 words. **Gold labels:** each review has a 1-5 star rating, which we map to sentiment labels (1-2 = negative, 3 = neutral, 4-5 = positive). After filtering and cleanup, we will sample 50,000 reviews approximately equally across the three classes (16,667 negative, 16,667 neutral, and 16,666 positive). All three methods will use the same stratified 80/10/10 train/validation/test split.

Team Member Contributions and Models

- **Longqi Sun:** Fine-tune a pretrained **BERT** model (bert-base-uncased) with a classification head for 3-class sequence classification, using standard cross-entropy loss on the balanced sample.
- **Quoc Cuong Dinh:** Classify reviews by **prompting a large language model** (Llama 3.1 8B Instruct, Hugging Face) with no task-specific training. Compare zero-shot with few-shot prompting using examples drawn only from the training set. The model will be instructed: *Return only one label: Positive, Neutral, or Negative.* The response will be normalized and mapped by exact match, with case-insensitive whole-word matching as a fallback. Also randomly sample 200 extra reviews per class (600 total) and have an AI (Gemini or Claude) label them through its API in small batches, without star ratings (the class split may be unequal).
- **Bhone Kyaw Si Hein:** Fine-tune the same **Llama 3.1 8B model with QLoRA** (the base model is loaded in 4-bit precision and only low-rank adapter weights are trained with the Hugging Face PEFT library) to generate one sentiment label. Training targets will contain only Positive, Neutral, or Negative, and inference will use the same extraction rule as the prompted method.
- **All members (shared equally):** filter reviews to 100-300 words; remove links, @ mentions or symbols, # symbols, and emojis; create the three rating-based labels; perform balanced sampling; create the shared split; and write the final report. Tokenization is handled automatically by each transformer model and is not a separate data-preparation step.

Evaluation Metrics

Because this is a sequence classification task, we will report **overall accuracy**, per-class **precision, recall, and F1**, and micro- and macro-averaged precision, recall, and F1 for each method. Macro-F1 will be the main comparison metric because it gives equal importance to all three sentiment classes. A confusion matrix will be used separately to analyze class-specific prediction errors; it will not be listed as a performance metric. All methods will be evaluated on the same held-out test set. **Experiment 2 (comparison against AI-assisted labeling):** all members will run their final models for inference on the 600 AI-labeled reviews and assess performance.

Computational Resources

Google Colab Pro (GPU, e.g., A100 or L4) for BERT fine-tuning, Llama prompting, and QLoRA fine-tuning of Llama 3.1 8B in 4-bit precision through Hugging Face. Both Llama methods will be evaluated using the extracted discrete labels, so logits or probability scores are not required.